SUGANESHWARAN R

Data Engineer

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PROFESSIONAL SUMMARY

Data Engineer with 3+ years of experience at Saturam building scalable data pipelines and reliable data platforms on AWS and Azure. Skilled in Python, Apache Airflow, Databricks (PySpark, Spark SQL, Delta Lake), Medallion architecture, and containerized delivery with Docker, Kubernetes, and Jenkins CI/CD. Built an Azure OpenAI-based synthetic data generator and a machine learning fraud-detection pipeline. Committed to delivering high-quality data solutions that support analytics, operational systems, and data-driven decision-making.

TECHNICAL SKILLS

Programming: Python, SQL, Shell Scripting

Data Engineering: Apache Airflow, Databricks (PySpark, Spark SQL, Delta Lake), AWS Glue, ETL/ELT, Medallion Architecture

Cloud: AWS (S3, Glue, Lambda, EC2, VPC, IAM, RDS), Azure (Azure OpenAI, Blob Storage, Key Vault, Microsoft Entra ID)

DevOps: Docker, Kubernetes, Jenkins, CI/CD, Terraform (Infrastructure as Code), Git

Databases: PostgreSQL, MySQL, Snowflake

Frameworks and Tools: Flask, React, Vite, Streamlit, AI-assisted development

AI and Blockchain: Azure OpenAI, machine learning model integration, Web3, Web3.py, Ethereum data extraction

WORK EXPERIENCE

Big Data Engineer 2023 – Present

Saturam – Advanced Data Engineering for Cloud & ML Ecosystem

- Architected and monitored complex data workflows in Apache Airflow with robust dependency handling, keeping production-grade pipelines reliable.
- Managed scalable environments using Docker and Kubernetes (Pods, Deployments, Services).
- Implemented CI/CD processes with Jenkins to automate deployments across AWS and Azure.
- Troubleshot frontend and backend environments to maintain stability and performance during high-traffic periods.
- Built and deployed the AI Synthetic Data Generator, an Azure OpenAI-based pilot for multi-table synthetic data (see Projects).

PROJECTS

AI Synthetic Data Generator (Azure)

Technologies: Azure OpenAI, Flask, Waitress, React, Vite, PostgreSQL, Azure Blob Storage, Azure Key Vault, Microsoft Entra ID

- Built and deployed an Azure-based pilot that lets authenticated users upload schemas, analyze table relationships with Azure OpenAI, generate realistic multi-table synthetic datasets, and export results as CSV, ZIP, or Excel files.
- Developed a React/Vite frontend and a Flask/Waitress backend with APIs for schema ingestion, parallel AI analysis, row-count recommendations, error-rate suggestions, LLM seed previews, full data generation, background jobs, and Blob-based downloads.
- Integrated PostgreSQL, Blob Storage, Key Vault, Microsoft Entra ID OAuth and Google SSO, managed identity, audit logging, SAS-token downloads, and Azure Marketplace SaaS fulfillment for secure authentication, secrets management, subscription handling, and scalable cloud deployment.

Automated AWS Data Lake (Medallion Architecture)
Technologies: Terraform, AWS (EC2, Glue, Lambda, VPC, RDS), Python, Streamlit

- Used Terraform (Infrastructure as Code) to provision a full-stack AWS environment including VPC, EC2 instances, and serverless Lambda functions.
- Developed a 3-tier Medallion data pipeline: ingested raw Kaggle datasets into RDS, processed Bronze and Silver layers with AWS Glue, and curated Gold aggregates for business intelligence.
- Built an interactive Streamlit dashboard to visualize Gold-layer insights in real time.
- Applied AI-driven natural language programming (vibe coding) to accelerate development cycles and improve code quality.

Blockchain ML Fraud Detection Platform
Technologies: Web3.py, Ethereum data, Flask, React, PostgreSQL, Kubernetes

- Built high-performance ETL pipelines to extract and process Ethereum blockchain data using asynchronous APIs.
- Integrated a machine learning model to detect high-risk crypto transactions, with automated fallback handling.
- Deployed with containerization, full logging, and monitoring, ensuring 99.5%+ uptime for real-time fraud analysis.

EDUCATION

Bachelor of Engineering (B.E.), Computer Science and Engineering 2020 – 2023
Government College of Engineering, Salem

Diploma in Computer Science and Engineering 2018 – 2020
Muthayammal Polytechnic Institution

CERTIFICATIONS AND TRAINING

- Databricks Fundamentals – Databricks Academy Accreditation
- Generative AI Fundamentals – Databricks Academy
- Generative AI Course – Simplilearn
- TCS NQT National Level Test – 77.80%
- Blockchain & Web 3.0 Workshop – EPNS SDE, MIT Chennai
- IoT and Blockchain: Challenges and Applications Workshop – GCE Salem

Certificate copies: https://drive.google.com/drive/folders/1FLE-2-0lfFPQpUHvOMDe-nQ0RWjxk9Lu?usp=sharing

LANGUAGES

English, Tamil